

Session 221: Communication Signal Processing and Coding

TIME 2:30 PM – 4:00 PM IST VENUE SP 1.20 (TVS Lab)



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- 221.1** **Hardware-Efficient FPGA Implementation of OTSM Transceiver for High-Mobility 6G Communications**
Vishnupriya P V, Sapta Girish Neelam
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- 221.2** **Iteratively reweighted algorithms for robust sparse recovery with L_p ($0 < p < 1$)-norm minimization**
Shweta Bharti, Samrat Mukhopadhyay, Govind Murmu
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- 221.3** **Coded SumComp: Linear Coding over Integer Rings for Improved SumComp Performance**
Vaishnavi Anekar, Anjana A M
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- 221.4** **A Novel Tunable Window for Signal Processing Applications**
Debanjan Bhattacharjee, Sujit Kumar Sahoo
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- 221.5** **PYNQ-Based Modular Architecture for Real-Time Stream-Cipher Video Encryption at the Edge**
Vanukuru Hari Rohit, D. Sreekiran, Raghavendra Kune, D. Mallikharjuna Rao, Novaline Jacob
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- 221.6** **GSM-Based Non-Orthogonal Multiple Access: Design and Evaluation**
Surjeet Kumar, Ashish Kant Shukla, Manav R Bhatnagar
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- 221.7** **Enhancing Power Efficiency in MIMO DFT-s-OFDM: A Relative Study of $\pi/2$ -BPSK with Tone Reservation**
Abhinaba Dey, Sreenath Ramanath
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- 221.8** **Molecular Communication with Finite-Length Transmitter in Time Dependent Diffusive Channel**
Neha Bhusari, Avinash G. Keskar, Nikhil Sharma, Prabhat Kumar Sharma
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- 221.9** **Generalized Partial Full-Pilot Zero-Forcing Combining for Distributed Massive MIMO**
Mohd Saif Ali Khan, Karthik RM, Samar Agnihotri
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- 221.10** **Signal Space Decoupled Stream Splitting for Low Latency Downlink MIMO WBANs**
Gunjit Arora, Satya Kumar Vankayala, Manoj Thomas
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- 221.11** **Low Complexity Digital Predistortion for Power Amplifiers Using Incremental PCA**
Girish Tripathi, Anindya Saha
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- 221.12** **Rank-Deficit Hybrid Adaptive Fulcrum Decoder for Low-Complexity Network Coding**
Sovanjyoti Giri, Satya Prakash
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- 221.13** **Evolutionary Optimization for Quantized RIS Phase Shift Design in MIMO Systems: Hyperparameter Analysis and SNR Performance**
Amit Shrivastava, Manish Kumar, Kunwar Pritiraj Rajput
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- 221.14** **Deep Unfolded AMP for Short Block-Length SPARC over AWGN Channels**
Sai Dinesh Kancharana, Sandeep VSV, Arun Pachai Kannu
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- 221.15** **Detection in Ambient Backscatter Communication System with Multi-Antenna Tag-Reader Pair with Correlated Ambient Source Signal**
Anuranjan Jha, Adarsh Patel

Session 222: Vision and Sensor Analytics

TIME 2:30 PM – 4:00 PM IST VENUE SP Building Side Foyer



SPCOM 2026

International Conference on
Signal Processing and Communications

222.1

Signal-driven Shipwreck Segmentation and Probabilistic Uncertainty Estimation with Sidescan Sonar for Autonomous Underwater Navigation

Lakshmi Annamalai

222.2

Vision-Powered Cleaning Mode Detection in Robot Vacuum Cleaners Using a Novel Smart Home Dataset

Sanket Gharatkar, Suhel Khan, Nagaraj Yendeti

222.3

NuSegFormer: Enhanced Cell Nuclei Segmentation Using a Hybrid UNet--Vision Transformer with Gated Multi-Head Attention

Aditi Damodar, Hitesh Tekchandani, Praveen Kumar Shukla, Anish Kumar Vishwakarma

222.4

On the Optimal Number of Grids for Differentially Private Non-Interactive K-Means Clustering

Gokularam Muthukrishnan, Anshoo Tandon

222.5

Are Wavelets More Reliable? Comparing The Topology of Image Generator Metrics

Manas Patil, Amit Tiwari, Abhijat Bharadwaj, Vikram Gadre

222.6

Domain-Adaptive Self-Supervised Pretraining for Bacterial Flagellar Motor Detection in Cryo-Electron Tomography

Adavi Jayanth, Madhu Oruganti

222.7

Enhancing Road Safety through Real-Time Unsupervised Anomaly Detection in Crash Barriers

Divya Varshney, Avnish Panwar, Indrajit Ghosh

222.8

Multi-Sensor Alignment for Weather Simulations

Samsad Alam, Devyani Lambhate, Aditya Mohan, Vishal Kumar, Vaibhav Katewa

222.9

SIADS: Innovation-Based Anomaly Detection in Streaming Communication Graphs

Tanish Verma

222.10

LS-PolypSeg: A Parameter-Efficient LoRA-Enhanced SAM3 Framework for Polyp Segmentation

Sanjana Jhansi Ganji, Srikanth Panigrahi, Kaushal Sambanna, Srinivas Rao Routhu

222.11

Tube Loss-based Detection of Anomalies in Electricity Consumption Time Series Data

Shruti Mehta, Pritam Anand, Yash Vasavada

222.12

Learning What to Trust: Confidence-Aware Curriculum Distillation for TinyML Vision

Satarupa Das, Soumen Garai, Rajrup Saha, Soumya Chatterjee, Suman Samui

Session 223: Wireless Resource Optimization

TIME 2:30 PM – 4:00 PM IST VENUE ECE 1.07



223.1 **Mobile Base Station Optimal Tour in Wide Area IoT Sensor Networks**
Sachin Kadam

223.2 **Robust Deep Reinforcement Learning for Task Scheduling in MEC-Enabled IoT Networks**
Vikky Masih, Arghyadip Roy

223.3 **DPAS: Demand-Pressure Adaptive Slicing for Real-Time 6G Network Resource Management**
Jaynil Panjwani, Neha Agrawal

223.4 **Ergodic Throughput in Coordinated Multi-point Downlink under Distributed Power Constraints**
Mahesh Nagulapalli, Mahendra Pratap Singh, Anil Kumar, Sibi Raj B. Pillai

223.5 **Mobility-Aware Offloading with SINR Prediction and Selective Replication in Fog-IoT Networks**
Rajasekhar Dasari, Satyajit Mohapatra, Sanjeet Kumar Nayak

223.6 **Decoding-Order-Aware Secrecy Rate Maximization in IRS-Assisted Untrusted NOMA Under Imperfect SIC**
Ankit Kumar, Kaneez Khatoon, Deepak Mishra, Ravikant Saini

223.7 **Reinforcement Learning-Based Decode and Forward UAV Relay Trajectory Optimization**
Aniket Subbanwar, Ojas Joshi, Amit Agarwal

223.8 **Measurement-Based Energy-Efficiency Optimization for On-demand Data Rate Provisioning in 5G RAN**
Ramagiddaiah Eediga, Preethi G V, Anand S V R, Chandra R. Murthy, Ryan Husbands

223.9 **Efficient Sampling for Pure Exploration Problems**
Dhinesh Chandran G, Srinivas Reddy Kota, Srikrishna Bhashyam

223.10 **Link Adaptation Using Joint-Thompson Sampling**
Vignatha Vinjam, Manjunath Kolavennu, Myna Vajha, Karthik Periyapattana Narayanaprasad

223.11 **Adaptive Scheduling for Low-Latency Coordination in Wi-Fi 8 Multi-AP Networks**
Hiya Shah

223.12 **Robust Base Station Placement in Agricultural IoT via Bayesian Optimization**
Gourav Prateek Sharma, Durgesh Singh, James Gross